J. Biophotonics **8**, No. 11–12, 993–1001 (2015) / **DOI** 10.1002/jbio.201500150

Journal of BIOPHOTONICS

FULL ARTICLE

Analysis of forward and backward Second Harmonic Generation images to probe the nanoscale structure of collagen within bone and cartilage

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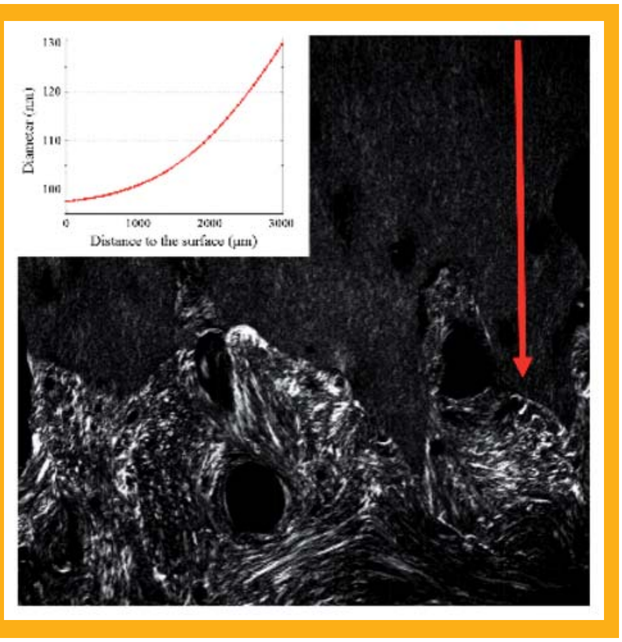
Received 4 May 2015, revised 17 July 2015, accepted 27 July 2015
Published online 10 September 2015

Key words: nonlinear microscopy, Second Harmonic Generation, forward/backward ratio, bone, cartilage

Collagen ultrastructure plays a central role in the function of a wide range of connective tissues. Studying collagen structure at the microscopic scale is therefore of considerable interest to understand the mechanisms of tissue pathologies. Here, we use second harmonic generation microscopy to characterize collagen structure within bone and articular cartilage in human knees. We analyze the intensity dependence on polarization and discuss the differences between Forward and Backward images in both tissues. Focusing on articular cartilage, we observe an increase in Forward/Backward ratio from the cartilage surface to the bone. Coupling these results to numerical simulations reveals the evolution of collagen fibril diameter and spatial organization as a function of depth within cartilage.



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1. Introduction

Connective tissues such as skin, tendon, bone or cornea are made of more than 90% of extracellular matrix (ECM) proteins, fibrillar collagens being by far the predominant components. Synthesized by cells as triple helix units, these collagens self-assemble *in vivo* and *in vitro* to form fibrils with a range of diameters and distributions [1, 2]. These structural proteins show a remarkable diversity in molecular and supramolecular organization [3, 4] and various collagen types lead to several macromolecular structures, specific to each tissue. Their hierarchical organization in the ECM plays a unique role in maintaining the structural integrity of most connective tissues and is responsible for the macroscale biomechanical and physical properties. Structural changes in the fibrillar matrix are associated with physiological processes during development, remodeling, aging, wound healing and disease [5–7]. In this study we focus on collagen type I and collagen type II. Collagen type I is the most common type of collagen in higher vertebrates and is the principal organic component of bones while collagen type II, which is less abundant, can be found almost exclusively in articular cartilage.

In recent years, Second Harmonic Generation (SHG) microscopy, a form of nonlinear laser scanning microscopy, has emerged as a powerful imaging technique to visualize the three-dimensional (3D) architecture of thick biological tissues [8–10]. Notably, SHG provides a non-invasive excitation method together with an intrinsic optical sectioning offering submicron 3D resolution, robust upon light scattering, and a high penetration depth in tissue. SHG is a coherent multiphoton process that does not involve electronic population transfer, preventing photobleaching and phototoxicity, and is specific for dense non centrosymmetric media such as fibrillar collagen [11–14]. Indeed, because of the tight alignment of harmonophores along the collagen triple helix and within fibrils, fibrillar collagen is highly anisotropic and the second harmonic generated is coherently amplified. Accordingly, fibrillar collagen exhibits strong endogenous SHG signals and allows highly contrasted images. Therefore, SHG microscopy appears as a sensitive structural probe of the macromolecular organization of collagen in both *ex vivo* and *in vivo* preparation [15–17]. To extract more structural information, Polarization-resolved Second Harmonic Generation (P-SHG) has been developed in the past years [18–23]. This technique uses different incident polarization angles to obtain the azimuthal distribution of the collagen fibrils inside the sample.

Since collagen fibril diameter in tissues (10–300 nm) is much smaller than the optical resolution, they are not resolved in the SHG images and it remains difficult to link the detected SHG signal to the collagen arrangement at the fibrillar scale. Indeed, because of their coherent nature, the mechanisms underlying SHG are more complex than those of fluorescence. In particular, the emitted signals retain the phase information from the excitation, and thus coherent interaction among harmonophores plays an important role in determining the overall intensity as well as the radiation direction [24]. Indeed, images acquired in the forward and backward direction exhibit dramatic structural differences because of the dissimilar phase-matching conditions, which offers the opportunity to obtain complementary information from the signal directionality [25–28].

In this manuscript, we investigate the structural differences between collagen type I from bone and collagen type II from articular cartilage. In particular, we use P-SHG and forward/backward (F/B) signal analysis to characterize the organization and bundling of the collagen meshwork in both tissues. Finally, we combine F/B measurements with a numerical model of the tissue to relate images ($\sim$100 μm) to the underlying collagen structure ($\sim$100 nm) to gain an insight in the collagen fibril diameter variations with depth from cartilage surface to the bone.

2. Methods

2.1 Samples preparation

4 samples were obtained from human tibial plateaus of patients undergoing above knee amputation. Each one was cut perpendicular to the surface of the knee and the sections were fixed in 10% formalin for 12 h. Samples were decalcified during 24 h using nitric acid and embedded in wax. Blocks were cut using Leica RM2135 microtome (Leica Microsystems, Germany) and 10 μm thick slices were deposited between a glass slide and a number 1 coverslip. Samples were then directly observed under SHG microscopy without further preparation or staining steps. A drop of water was placed on the top of the coverslip to prevent back reflections which would affect the ratio between the SHG intensities in the forward and backward directions.

2.2 Experimental setup

A titanium:sapphire oscillator (Tsunami, Spectra Physics, Santa Clara, USA) was used to generate $\sim$150 fs laser pulses with a central wavelength of 810 nm and a 80 MHz repetition rate. The average excitation power was adjusted using a half-wave plate and a Glan Thompson polarizer. Images were acquired with a laser scanning microscope (Till Photonics GmbH, Munich, Germany) based on galvanometric mirrors. The focusing objective was an Olympus UplanSApo 20X air immersion microscope objective with a numerical aperture of 0.75. To achieve optimal spatial resolution of typically 600 nm (FWHM), the beam size was increased using a telescope as to entirely cover the back aperture of the microscope objective. Mechanical and piezoelectric motors, for coarse and fine adjustments respectively, were used to vertically move the focal spot in the sample.

SHG signals were collected both in forward and backward direction and detected using photomultiplier tubes (model R6357, Hamamatsu Corporation, New Jersey, USA) set at 600 V. Appropriate filters are used to reject the excitation light (380–700 nm, SEMRock, Rochester, NY, USA) and select the wavelength of interest (405 ± 10 nm, SEMRock, Rochester, NY, USA). In the forward direction, the signal was collected using an objective with a numerical aperture of 0.8 (40X, Olympus, Japan) whereas in the backward direction, the SHG light was collected through the microscope objective and reflected by a long pass dichroic mirror (FF735-Di01-25×36, SEMRock, Rochester, NY, USA) towards the PMT.

Polarization-resolved imaging was achieved by tuning the pump beam polarization using a half-wave plate (Thorlabs, Newton, NJ, USA) placed before the galvanometric mirrors. A Berek compensator (Newport, Irvine, CA, USA) was positioned immediately after the half-wave plate to pre-compensate the ellipticity induced by the dichroic mirror inside the microscope. For each region of interest (ROI), images were acquired with a polarization varying from 0 to 180° with 10° steps.

2.3 Numerical simulations

Numerical simulations were performed to calculate the forward and backward SHG intensity generated by an assembly of fibrils as a function of the fibrils diameter. This allows us to perform a quantitative analysis of SHG images as a function of the collagenous structure within tissue. For a complete description of the calculation see [29, 30].

Samples were modeled as an assembly of infinitely long cylinders, lying in the focal plane, with the same diameter (d). The nonlinear response of every cylinder is described through a constant scalar nonlinear susceptibility $\chi^{(2)}$ [17], with a randomly assigned sign from one fibril to another. In this model, all cylinders are aligned in the laser polarization direction which provides a maximum in both forward and backward SHG intensity. The model tissue is characterized by the filling ratio (ρ) indicating the volume fraction of fibrils within the focal volume, and a constant refractive index of 1.4. The excitation is modeled by a Gaussian beam at 810 nm with a waist of 1.25 µm. We compute both forward and backward SHG signal from the simulated tissue. Briefly, the SHG electric field generated by individual fibrils is expressed in terms of Hertz vector, satisfying Helmholtz equation. This equation is solved in the far-field using a vectorial modification of the scalar Green's function approach [31]. In this way, the nonlinear polarization is multiplied by the Green's function determined and integrated over the entire volume of each cylinder to evaluate the SHG field emitted by individual fibril in the far field zone. Finally, we coherently add every cylinder contribution to evaluate the total second harmonic field generated by the tissue in the focal volume.

It is worth noting that the highly disorganized collagen meshwork in articular cartilage does not present the spacing periodicity required to give rise to a significant coherent contribution of the forward reflected SHG to the backward emission. In addition, the quite low difference between refractive indices results in low Fresnel reflection coefficients. Finally, the thin slices used in this study (10 µm) does not scatter the light enough to give rise to a significant backscattering of the forward SHG signal [26]. Therefore, we neglected the contribution of both reflected and backscattered forward SHG signal in the numerical simulations.

The forward and backward SHG signals in a model tissue are entirely described through d, ρ and the ratio of fibril with positive $\chi^{(2)}$ over the total number of fibrils (f ratio [30]). As the simulated SHG depends on the local structure of tissue, we compute the average intensity from 1000 randomly generated modeled tissues for each value of ρ and d.

3. Results and discussion

3.1 Comparison between forward and backward SHG images

Figure 1 shows typical forward (Figure 1a) and backward (Figure 1b) SHG images of cartilage-on-bone samples from the human knee. In the top part of both images, we observe collagen type II from articular cartilage while the bottom shows collagen type I from bone. This highlights the different structural arrangements of type I and II collagens within these tissues. We first observe that the SHG signal from cartilage, both in the forward and backward direction, seems to be more uniform than from the bone. In addition, in the forward direction, SHG intensity from type I collagen in the bone appears to

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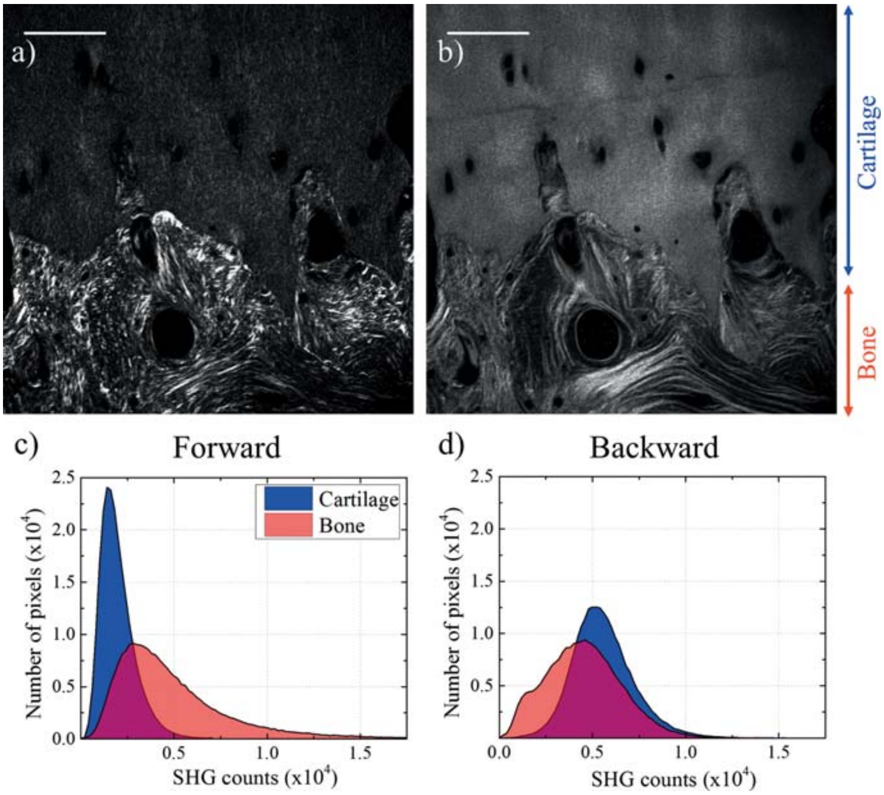

Figure 1 Typical SHG images of collagen type I from bone (bottom of the image) and type II from articular cartilage (top of the image) acquired in (**a**) forward and (**b**) backward direction. Scale bars: 100 μm. Histograms of the collagen SHG intensity distribution in the bone and the articular cartilage in (**c**) forward and (**d**) backward direction.

be significantly higher than the intensity of type II collagen from cartilage while in the backward direction, the intensity seems to be of the same order for both types of collagen.

Figure 1c and d show the histograms of the SHG intensity from a $100 \times 100\ \mu m^2$ region of interest (ROI) within bone and articular cartilage. Importantly, the ROI selected in the bone is one that has few cells. The size of the ROI was chosen as to ensure the local flatness of the sample and prevent intensity variations due to out of focus excitation. In addition, since we used different excitation powers to acquire forward and backward images, we normalized the image by dividing the intensity by the square of the laser power. Histograms confirm that the SHG intensity in the backward direction is similar for both types of collagen, the two distributions having a similar average number of counts, whereas the distribution of intensity in the forward direction changes significantly depending on the tissue. Indeed, in the cartilage the forward signal is relatively uniform, as in the backward direction. On the contrary, the bone exhibits a higher average number of counts and the signal is distributed over a far broader range and presents, notably, pixels with significantly higher intensity (over 10^4 counts).

The different behaviors exhibited by the forward and backward intensities can be understood by considering the phase matching in the sample and highlights the different fibrillar organizations in the two tissues. Indeed, the SHG intensity is directly affected by the collagen structure at the scale of the coherence length $l_c = \pi/\Delta k$, where Δk is the phase mismatch. Due to the geometry of detection, the coherence length in the backward direction (~70 nm) [29] is smaller than the fibril diameter (typically 100 nm) [32], and the organization and bundling of fibrils does not influence significantly the backward signal. On the contrary, in the forward direction, the coherence length is approximately 1 μm, considerably larger than the fibril diameter. Therefore, the forward signal results from an averaging of the size, the organization and the orientation of fibrils within the focal volume, which gives rise to wide variations of intensity not visible in backward detection. In particular, the high intensity pixels in the bone (top of Figure 1a) reveal the presence of bundles of collagen type I fibrils with the same polarity. This bundling results in strong forward SHG signal, due to the coherent addition of all the fibrillar contributions, whereas the backward intensity remains almost unaffected [24] because the bundles diameter is larger than the backward coherence length. This effect is not visible in the cartilage (Figure 1a and b) which indicates a low level of bundling in this tissue and a quasi-random organization of collagen fibrils at the micrometer scale.

3.2 Polarization-resolved SHG in bone and cartilage

Figure 2a and b show respectively the forward and backward images of type I collagen within the bone. Rotating the polarization of the incident laser beam allowed us to measure the polarimetric response of collagen fibrils in different regions of the tissue. As previously demonstrated [33–35], the SHG signals from collagen fibrils are higher when the polarization is parallel to the local direction of the fibrils, revealing the general alignment of the fibrils in the region of interest. Therefore, the general fibrils' orientation over the field of view can be observed on the corresponding polarimetric plot (Figure 2c) indicating the SHG intensity detected as a function of the incident polarization. For every polarization angle, the SHG intensity is thresholded, to exclude pixels without SHG signal, and averaged over sub-ROIs ($2 \times 2\ \mu m^2$). The error bars on the plot correspond to the standard deviation of the distribution. Finally, to quantify locally the degree of alignment of the tissue, we calculated, in every sub-region, the anisotropy parameter (r):

$$r = \frac{I_{\mathrm{par}} - I_{\mathrm{perp}}}{I_{\mathrm{par}} + 2I_{\mathrm{perp}}}$$

as previously proposed by Campagnola et al. [36]. Figure 2d displays a map of anisotropy where r is color coded in every sub-region, a hotter color indicating a higher degree of alignment.

Figure 2e and f present respectively forward and backward SHG images of type II collagen fibrils within articular cartilage. Figure 2g displays the SHG intensity, averaged over $2 \times 2\ \mu m^2$ sub-ROIs, as a function of the polarization angle. It is worth noting that, in the backward direction, despite the speckle like aspect of the image, the polarimetric measurements exhibit the same behavior as in the forward direction.

The different intensities and incident polarization dependencies are a signature of the different meshwork structures, at submicron scale, of collagen I and II. In the bone, the collagen fibrils are well aligned at the micron scale as revealed by the high anisotropy ratio (Figure 2d). However, the large distribution and the high standard deviations observable in Figure 2c denote a wide variation of the main

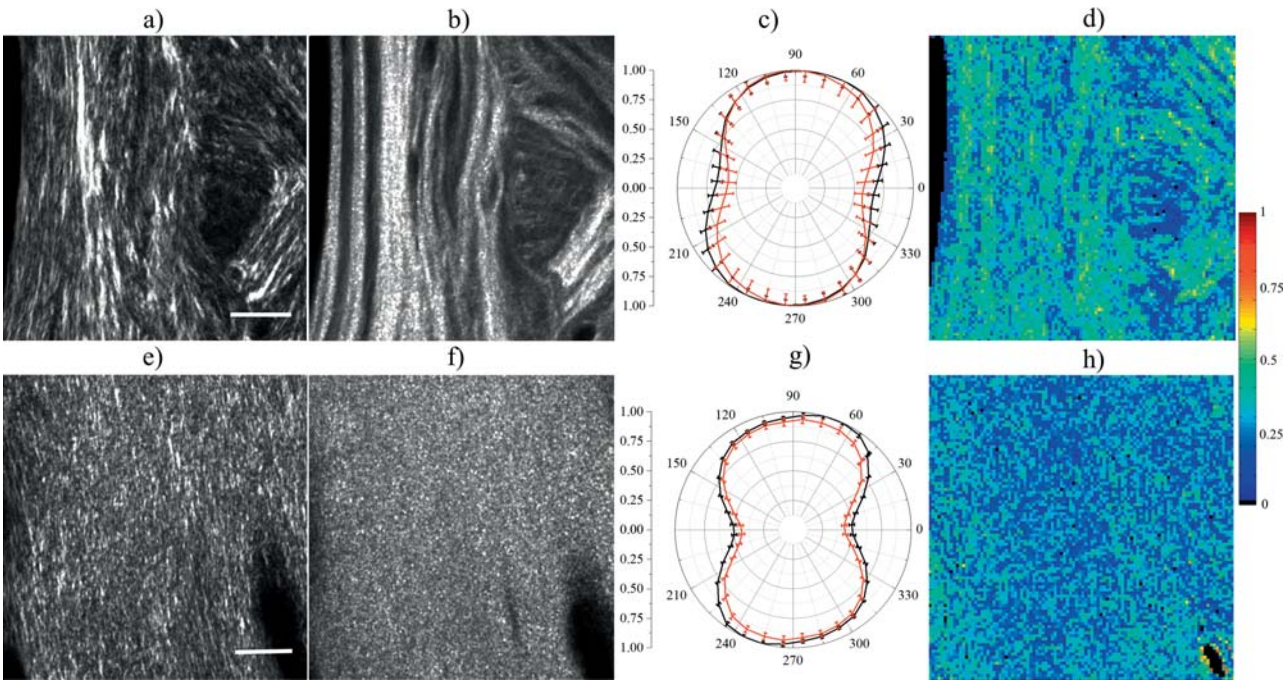

Figure 2 SHG images of (**a–b**) bone and (**e–f**) articular cartilage in forward and backward direction respectively. Scale bars: 20 μm. (**c**) and (**g**) SHG intensity of sub-regions, in bone and cartilage respectively, as a function of the incident polarization angle. The whole field of view has been divided in sub-regions ($2 \times 2\ \mu m^2$) and for every polarization angle, the SHG intensity has been averaged over all sub-regions in the forward (black line) and backward direction (red line). Intensity in both directions has been normalized so that the values range between 0 and 1. The error bars indicate the standard deviation of the intensity over the 2500 sub-regions. (**d**) and (**h**) map of the anisotropy ratio (**r**) calculated in the same sub-regions. The anisotropy is color coded revealing a locally higher degree of alignment in bone than cartilage.

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direction from one sub-ROI to another. Together with the high SHG intensity modulation, this indicates that collagen I fibrils are packed into bundles [37] at the micron scale, in which fibrils present both the same orientation and the same polarity. At a larger scale (~10 microns), the orientation of these bundles vary widely over the field of view, as observable in the image. This specific structure provides a high resistance to tensile forces which is consistent with the physiological function of bones. On the contrary, in articular cartilage, the low and relatively uniform anisotropy ratio indicates that collagen II is organized, at the micrometer scale, into a quasi-random meshwork of fibrils. This results in a low and uniform forward SHG intensity due to partially destructive interferences. However, at larger scale, the narrow polarization response distribution and the very small standard deviations observable in Figure 2g indicates a quasi-constant main orientation of collagen II fibrils from one sub-ROI (2×2 μm^2) to another, over the field of view (100×100 μm^2), the fibrils being perpendicular to the cartilage surface, which is consistent with the well-known arcade structure of collagen fibrils within this tissue [38, 39]. This quasi-random meshwork structure provides a high resistance to osmotic swelling and pressure which matches the function of articular cartilage [40].

3.3 Measurement of the fibrils diameter in cartilage using F/B ratio

Figure 3a and b show the forward over backward ratio, calculated pixel by pixel, in cartilage and bone respectively. To compensate for the differences in the detection efficiency between the forward and backward detection paths, the F/B ratio was first measured using two photon excited fluorescence from a solution of Coumarin 440 and using the exact same filters as for SHG images. Since fluorescence signals are inherently isotropic, this measurement allowed to calibrate the F/B ratio measured in the microscope. Therefore, we corrected the values measured in the cartilage accordingly, which allowed to extract information about the emission direction.

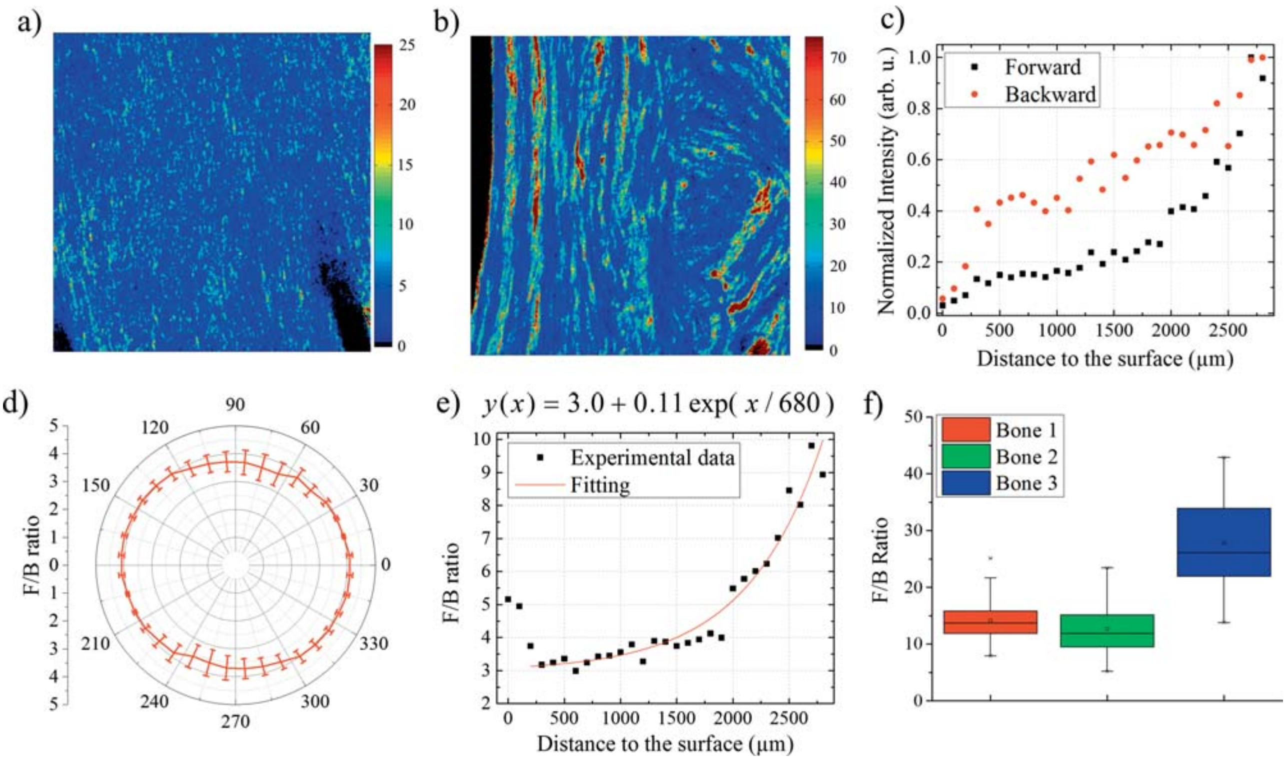

Figure 3 Images of the forward over backward ratio in (**a**) articular cartilage and (**b**) bone. (**c**) Normalized forward (black squares) and backward (red circles) SHG intensity fron articular cartilage as a function of the distance from the cartilage surface. Note that the backward signal seems higher than the forward only because of the normalization. (**d**) F/B ratio as a function of incident polarization angle at 500 μm depth from cartilage surface. (**e**) F/B ratio of the SHG intensity from articular cartilage as a function of the distance from the cartilage surface. The origin is the surface of the cartilage and the final position (3000 μm approximately) corresponds to the limit between bone and cartilage. An exponential fit (red line) has been performed over the experimental data, with the exception of the three first points. (**f**) Statistial distribution of the F/B ratio in three different ROIs in the bone. Each area of 100×100 μm^2 has been divided in 10×10 μm^2 regions where the F/B ratio is calculated.

Focusing on articular cartilage, Figure 3c shows the forward (black squares) and backward (red circles) SHG intensities as a function of distance from the surface to the bone. Because we used thin slices cut perpendicularly to the surface of the knee, the whole depth of the cartilage structure was available for imaging under the same conditions without attenuation, scattering or birefringence. To maximize the signal-to-noise ratio, we illuminated the sample with the linear polarization orthogonal to the cartilage surface, which correspond to the highest SHG signal in our polarimetric plots (see Figure 2g). The origin (0 µm) represents the surface of the cartilage and each point gives the mean SHG intensity over a $100 \times 100\ \mu m^2$ field of view. This field of view is then moved by 100 µm to the next point until the junction between bone and cartilage is reached (see Figure 1a and b), which correspond to a depth of around 3000 µm, depending on the sample. We observed a gradual increase of both the forward and backward SHG signals in cartilage when moving the field of view from the surface to the bone junction. In particular, in the first 300 µm after the surface, we observe a faster increase in both forward and backward SHG. This is attributed to the arcade structure of cartilage [38–40]. Indeed, the three closest fields of view from the surface correspond to the superficial and transitional layers where the collagen fibrils are very organized but curve to become parallel to the surface. This modification of the azimuthal angle implies that, in the three first images, the polarization is almost perpendicular to the fibrils, which results in both lower forward and backward signals without affecting the F/B ratio, as observable on Figure 3d showing the F/B ratio in cartilage as a function of the incident polarization.

Figure 3e displays the forward over backward ratio (F/B) as a function of the distance to the surface. It appears that the forward signal increases at a higher rate than the backward signal as we approach the junction to the bone, which leads to a gradual increase in F/B from the surface to the bone. Again, the only exception is the first 300 µm after the surface, where we observe a higher drop in the backward intensity causing an increase in the F/B ratio. This cannot be explained by an azimuthal angle which affects both the forward and backward SHG signals but not their ratio. We attribute the high F/B ratio at the cartilage surface to the change of the global meshwork orientation in these region, where a significant part of the collagen fibrils have a polar angle [38, 40, 41], which has been shown [29] to cause a high decrease in the backward signal but a small variation of the forward intensity, as observed in Figure 3c.

Figure 3f shows the average F/B ratio for type I collagen from bone. Three sets of data have been acquired from three different ROIs ($100 \times 100\ \mu m^2$) in the sample, each of them being thereafter divided in a grid of $10 \times 10\ \mu m^2$ regions where the F/B ratio is measured. The graph displays the distribution of F/B in each region for the 3 ROI. We observe high variations of the F/B ratio in the bone, varying from 5 to 45. This is probably caused by the bundling of type I collagen within the bone which increases the forward SHG intensity but does not change significantly the backward SHG intensity, again because of the different phase matching conditions.

It is worth noting that the F/B ratio measured here does not translate easily to *in situ* measurements on thick tissue, in which the backward SHG signal is combined with backscattered forward SHG. However, several approaches have been proposed to decouple these two contributions [26, 42] which would extend the potential of this analysis to perform *in situ* studies within thick biological tissues.

To further investigate the relation between F/B ratio, at the micron-scale, with the underlying col-

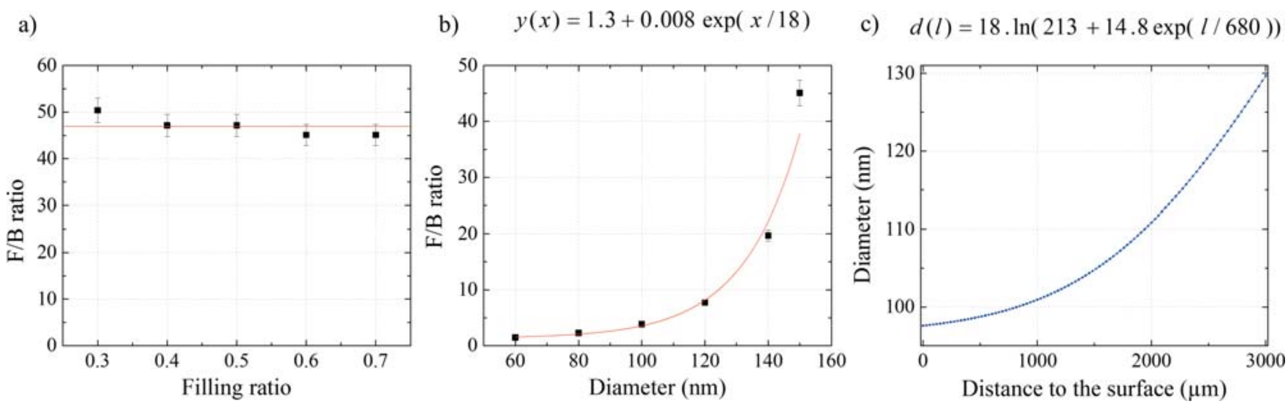

Figure 4 Simulated F/B ratio as a function of (**a**) the filling ratio for d = 150 nm and (**b**) the fibril diameter for ρ = 60%. Error bars indicates the average signals over 1000 randomly generated model tissues for every values of ρ and d. (**c**) Evolution of fibrils diameter within articular cartilage as a function of the distance to the surface as predicted from the F/B measurements (Figure 3e) and the numerical model.

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lagen structure, we developed a numerical approach to calculate both forward and backward SHG signal in tissue. Interestingly, the simulated forward and backward intensities were both found to depend linearly on the filling ratio [29] and therefore, the F/B ratio remains unaffected by the variation of ρ, as shown in Figure 4a. Figure 4b shows the evolution of the F/B ratio as a function of the fibrils diameter.

The increase in both forward and backward signals observable in Figure 3c indicates an increasing filling ratio. However, as this parameter does not affect the F/B ratio, the behavior of Figure 3e, where the F/B ratio increases from 3.2 near the cartilage's surface to 9.8 at the junction between bone and cartilage, indicates that the diameter of the fibrils is also increasing from the surface to the bone. From these values of F/B ratio, the numerical simulations (Figure 4c) allows to predict an increase in collagen fibrils diameter from 85nm at the surface to 125 nm at the bone-cartilage junction. These observations are consistent with the literature [39,43] in both values and variations as it is know that the collagen II fibrils in cartilage progressively become thicker from the surface to the bone. In the bone, the F/B ratio is generally higher (around 15) but presents a very wide dispersion, from 5 to 45, which indicates the presence of large fibrils ($\varnothing \sim 150$ nm) or bundles of fibrils sharing the same polarity [29].

4. Conclusion

In this paper, we report the use of Second Harmonic Generation microscopy to obtain structural information about the type I and II collagen scaffold within human bone and cartilage. In particular we performed Polarization-resolved SHG and analyzed the differences between images acquired in forward and backward direction. In the cartilage, the SHG signal is nearly uniform throughout the collagen meshwork, revealing that collagen fibrils are distributed quasi-randomly within the focal volume of the multiphoton microscope. This fits with a function of entrapment in which collagen provides a tensile force balance to high swelling aggrecan during load carriage. In contrast, the large variations observed in the SHG images of the bone suggest that the fibrils are packed into bundles with a constant orientation and polarity at the micrometer scale, indicating a more conventional fiber reinforcement role. Finally, in cartilage, we observed an increase of the F/B ratio from the surface to the bone. Analyzing this result in respect to numerical simulations, we demonstrated that the F/B ratio behavior follows the evolution of the collagen fibrils diameter and therefore provides information about the collagen structure at sub-resolution scale. This work illustrates the relevance of SHG microscopy to study the nanoscale architecture of connective tissues and presents a strong potential for a wide range of applications such as the investigation of remodeling processes, the characterization of interactions between cells and collagen scaffolds and the study of fibrillation mechanisms in pathological tissues.

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